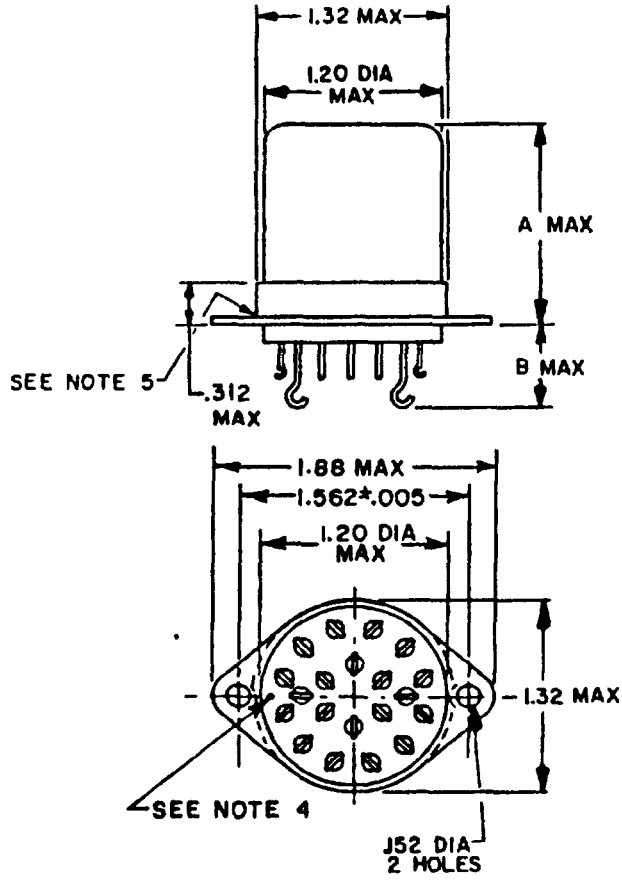
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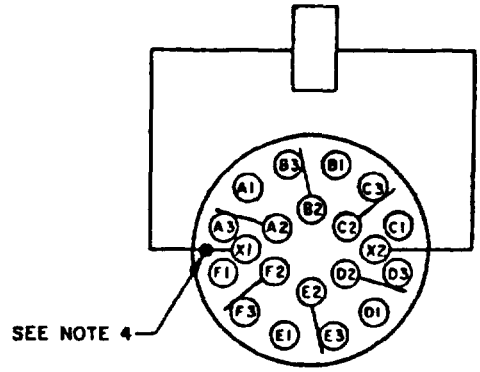
RELAYS, ELECTRICAL, HERMETICALLY SEALED, 6PDT, LOW LEVEL TO 2 AMPERES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-R-5757.



Inches	mm	Inches	mm
.005	0.13	1.32	33.5
.312	7.92	1.562	39.67
1.20	30.5	1.88	47.8



SCHEMATIC DIAGRAM
DEENERGIZED POSITION
BOTTOM VIEW

Dash numbers	A max	B max
-021, -027, -103	1.38 (35.1)	.56 (14.2)
-028	1.10 (27.9)	.84 (21.3)

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ± 010 (0 25 mm)
3. Metric equivalents are given for general information only.
4. A colored line or dot shall be placed in the position shown, or optionally, the adjacent terminal in a clockwise direction around the outermost terminal circle, shall have a blue bead.
5. Mounting screw head clearance shall be provided so that the relay may be mounted using a round head machine screw having 250 (6 35 mm) nominal head diameter; also the mounting surface shall be flat, with a minimum .276 (7.01 mm) diameter, and concentric with the mounting hole.

FIGURE 1 Relay - solder lug configuration (M5757/1-021, -103, -027, and -028).

(H) denotes changes

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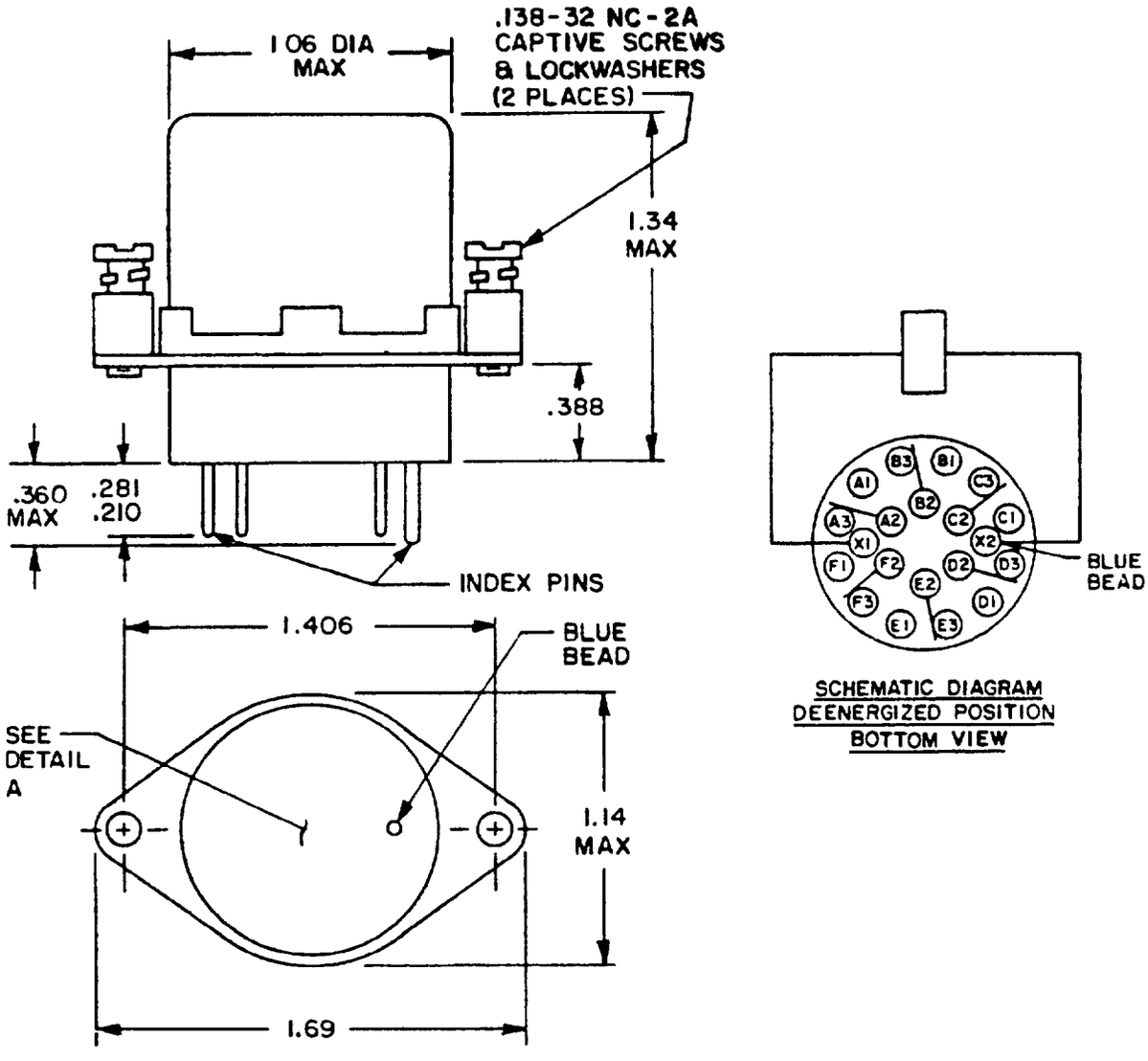
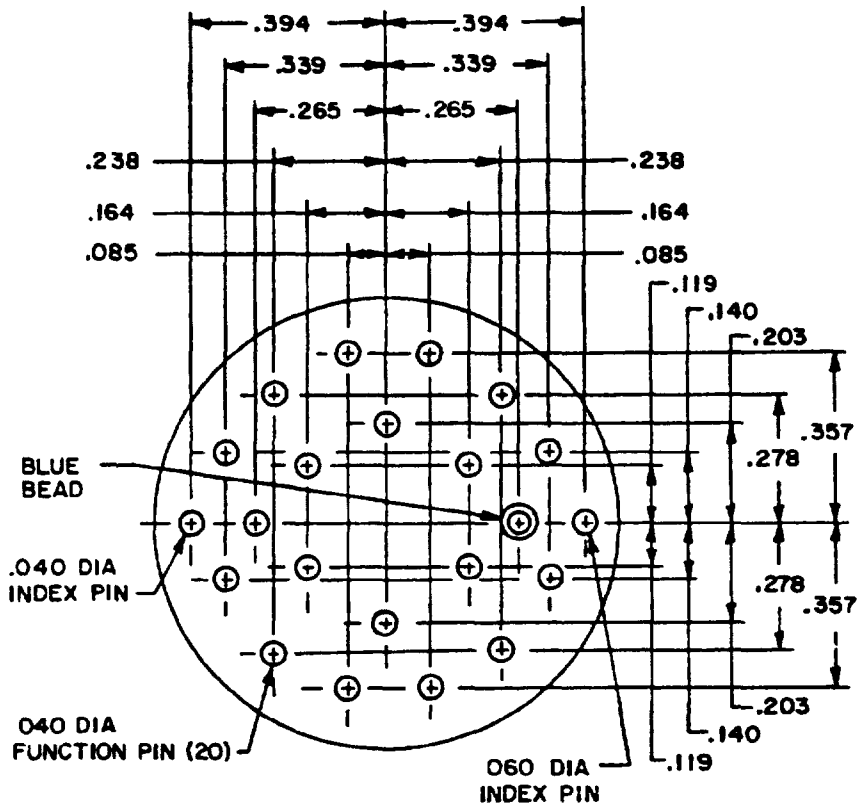


FIGURE 2 Relay - socket pin configuration (M5757/1-026)

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DETAIL A

nches	mm	Inches	mm	Inches	mm	Inches	mm
.040	1.02	.164	4.17	.281	7.14	1.06	26.9
.060	1.52	.203	5.16	.339	8.61	1.14	29.0
.085	2.16	.210	5.33	.357	9.07	1.34	34.0
.119	3.02	.238	6.05	.360	9.14	1 406	35.71
.138	3 51	.265	6.73	.388	9.86	1.69	42.9
.140	3.56	.278	7.06	.394	10.01		

NOTES
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
3. Metric equivalents are given for general information only.
4. (H) All active electrical terminals shall be gold plated per MIL-G-45204, type II, class I, 0.00005 inch (50 microinches) *minimum*. Gold plating of index pins is optional. Underplating, nickel 50 to 150 microinches.

FIGURE 2 Relay - socket pin configuration (M5757/1-026) - Continued.

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REQUIREMENTS:

Contact data:

	High-level characteristics (M5757/1-103, -027, and -028)	Low-level characteristics (M5757/1-021, -026, -027, and -028)
Configuration	6PDT	6PDT
Arrangement	6 form C	6 form C
Load ratings (relay case grounded):		
Resistive	2 amperes at 28 V dc	30 microamperes at 30 mV, dc to peak ac
Minimum current	Applicable -027 and -028	Not applicable
Contact resistance or voltage drop:		
Rated life:		
Before	0.05 ohm max	0.05 ohm max
During	10 percent of open circuit voltage max	100 ohms max
After	0.1 ohm max	0.15 ohm max
(H) Contact bounce	1 ms max (-103, test at rated resistive load) 2 ms max (-027 and -028)	2 ms max (-021, -027, -028) 5 ms max (-026)
Overload:		
Resistive	Two times rated current	Not applicable

Coil Data:

Duty rating	Continuous
Maximum voltage	32 V dc
Nominal voltage	26.5 V dc
Pickup voltage	18 V dc max over temperature range
Dropout voltage	1.5 V dc min, 14 V dc max over temperature range
(H) Coil resistance	180 ohms min, 220 ohms max at 25°C, except for -026 the coil resistance shall be 260 ohms max at 25°C
Operate time	15 ms max
Release time	15 ms max

Electrical data:

Insulation resistance - - - - 1,000 megohms min, except the resistance between coil and case at high temperature shall be 500 megohms or greater.

Dielectric withstanding voltage:

	Sea level V rms (60 Hz)	Altitude V rms (60 Hz)
Between case, frame, or enclosure and all contacts both in the energized and deenergized position	1,000	350 - All terminals to case
Between case, frame, or enclosure and coil	1,000	
Between all contacts and coil	1,000	
Between open contacts in the energized and deenergized positions	500	
Between contact poles	1,000	

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Environmental data:

Temperature range	Symbol B (-65°C to +125°C)
Internal moisture	Applicable
Vibration	Symbol 3 (15 g's, 10 to 2,000 Hz) and Symbol 4 (20 g's, 10 to 2,000 Hz)
Acceleration	Applicable
Shock	Symbol 3 (100 g's)

Physical data:

Terminal strength	5 ±0.5 pound pull
Sealed by welding	Optional
Dimensions and configuration	See figures 1 and 2
Terminations	See figures 1 and 2
Weight	0.27 pound max

Life test requirements:

Ⓗ High level	100,000 cycles
Ⓗ Low level	100,000 cycles

Ⓗ Part number: M5757/1-(applicable dash number from table III).

QUALITY ASSURANCE:

Group A

1. Subgroup 1: Conditioning (applicable to -021, -026, -027, and -028 relays only)

2. Subgroup 2: 100 percent.

3. Dielectric withstanding voltage:

   a. Tests to be conducted at sea level rating only.

   b. Duration of application: 5-10 seconds at a 10 percent increase in the dielectric strength voltage.

Qualification inspection: See table I or II as applicable.

Ⓗ TABLE I. Qualification inspection and sample size.

Part number M5757/1-	Number of sample units
021	12 units plus 1 open unit Qualification inspection as applicable
103	12 units plus 1 open unit Qualification inspection as applicable (except contact bounce shall be performed at rated resistive load)
026	12 units plus 1 open unit Qualification inspection as applicable
027 and 028	20 units plus 1 open unit Qualification inspection as applicable

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Ⓗ TABLE II. Extension of qualification and sample size.

Part number M5757/1-		Test	Number of sample units
Qualified	Extension to		
021	026	Visual and mechanical	1
021	103	High level life Electrical characteristics 1/	4
021	027 and 028	Visual and mechanical	1
		High level life	4
		Minimum current	4
103	021 and 026	Low level life	4
026	021	Visual and mechanical	1
026	103	Visual and mechanical	1
		High level life Electrical characteristics 1/	4
027	021 and 028	None required	-
	103	Electrical characteristics 1/	2

1/ Contact bounce shall be performed at rated resistive load.

Ⓗ Qualification by similarity If the relay case, frame, or enclosure contains integral electromagnetic relays (mounting means excepted) currently listed on the qualified products list of MIL-R-39016, reduced testing shall consist of subjecting 4 sample units to the shock and vibration requirements of this specification sheet. Post tests shall include insulation resistance, dielectric withstanding voltage and electrical characteristics and seal. One unsealed unit shall be submitted to the qualifying activity.

TABLE III. Part number and applicable characteristics.

Part number M5757/1-	Vibration grade
Low level	
021	4
026	3
High level	
103 1/	3
Low to High level	
027 2/	4
028 2/	4

1/ Not for new design after 5 March 1969.
2/ Each relay possesses high level and low level capabilities; however, relays previously tested at high level loads shall not be used for low level applications.

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Ⓗ Retention of qualification (Applicable only to relay assemblies which contain integral relays): 4 units, M57575/1-026, Group B tests as specified in MIL-R-5757 plus shock and vibration in three planes, electrical characteristics, and gross leak (seal).

SUPERSESSION DATA: See tables IV and V.

Ⓗ TABLE IV. Supersession data by part number.

Canceled part number M5757/1-	Replacement part number M5757/1-
001	027
002	027
003	027
004	No replacement
005	No replacement
006	No replacement
007	027
008	027
009	027
010	No replacement
011	No replacement
012	No replacement
013	021 or 027
014	021 or 027
015	021 or 027
016	No replacement
017	No replacement
018	No replacement
019	021 or 027 1/
020	021 or 027 1/
022	No replacement
023	No replacement
024	No replacement
025	027
101	103 or 027 1/
102	103 or 027 1/

1/ For Government logistics support M5757/1-027 supersedes -021 and -103.

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TABLE V. Supersession data by FSCM.

Superseded		Superseding
FSCM	Part number	Part number M5757/1-
80063	SM-C-415280	026

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AT, MI
Navy - SH
Air Force - 11, 17, 99
DLA - ES

User activities
Army - AR, AV
Navy - MC
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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